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(54) **CRYSTALLINE FORMS OF VAMOROLONE**

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(57) **ABSTRACT**

Provided are certain polymorphic forms of vamorolone as well as pharmaceutical compositions and methods for their use.

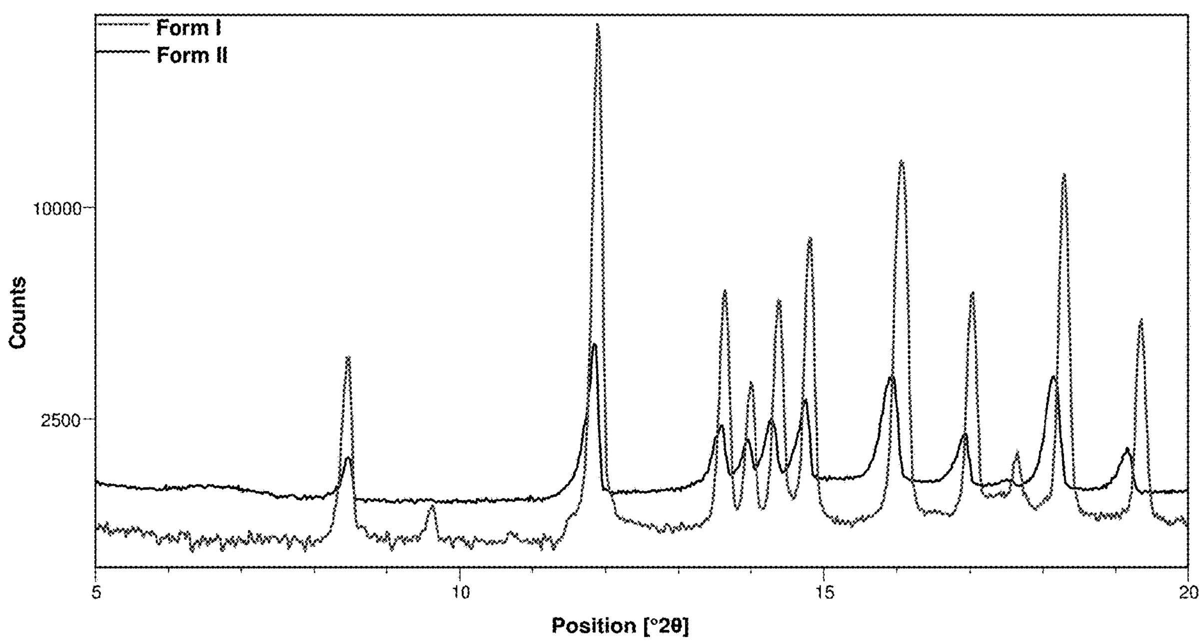


Figure 1

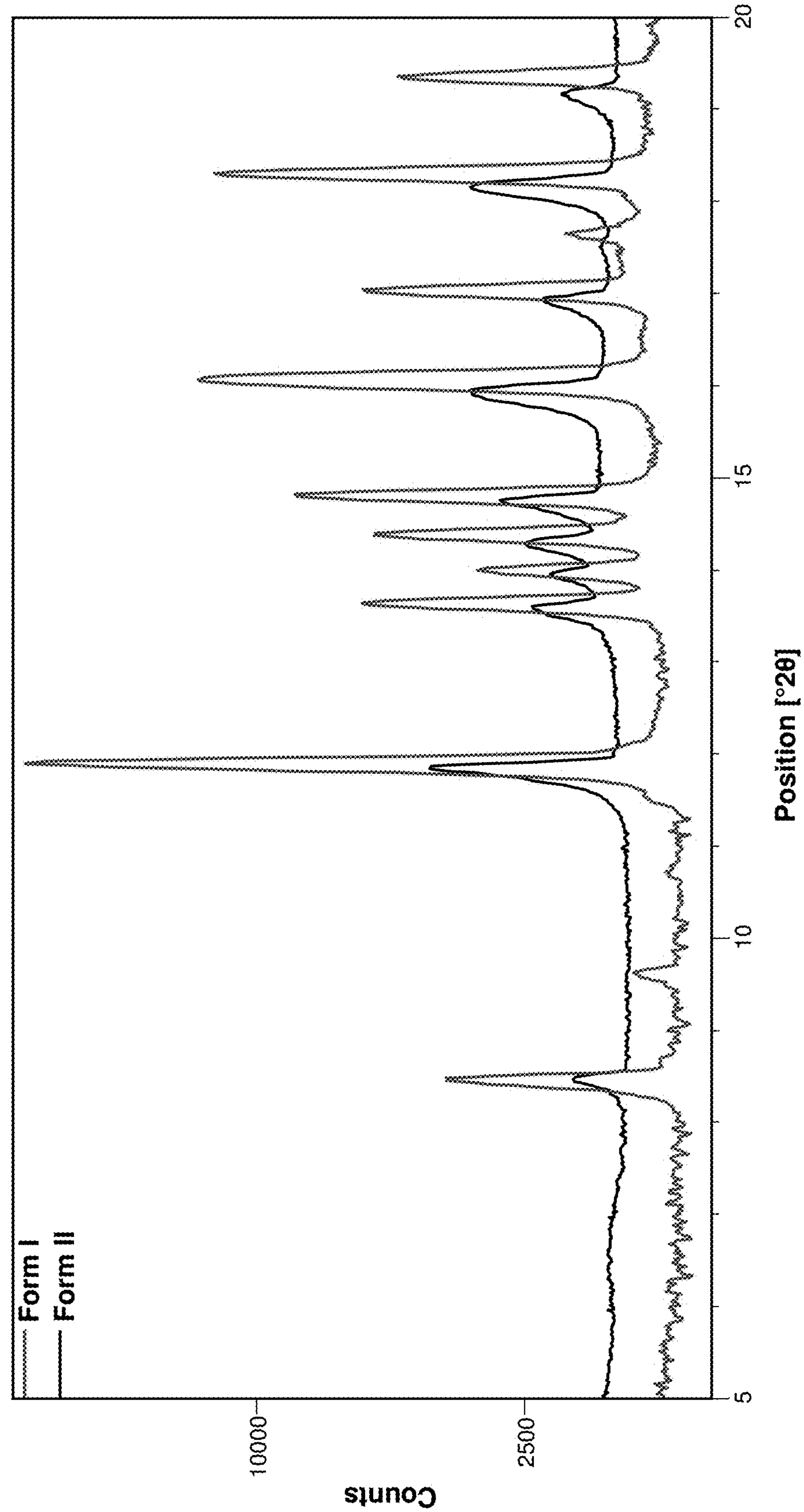


Figure 2

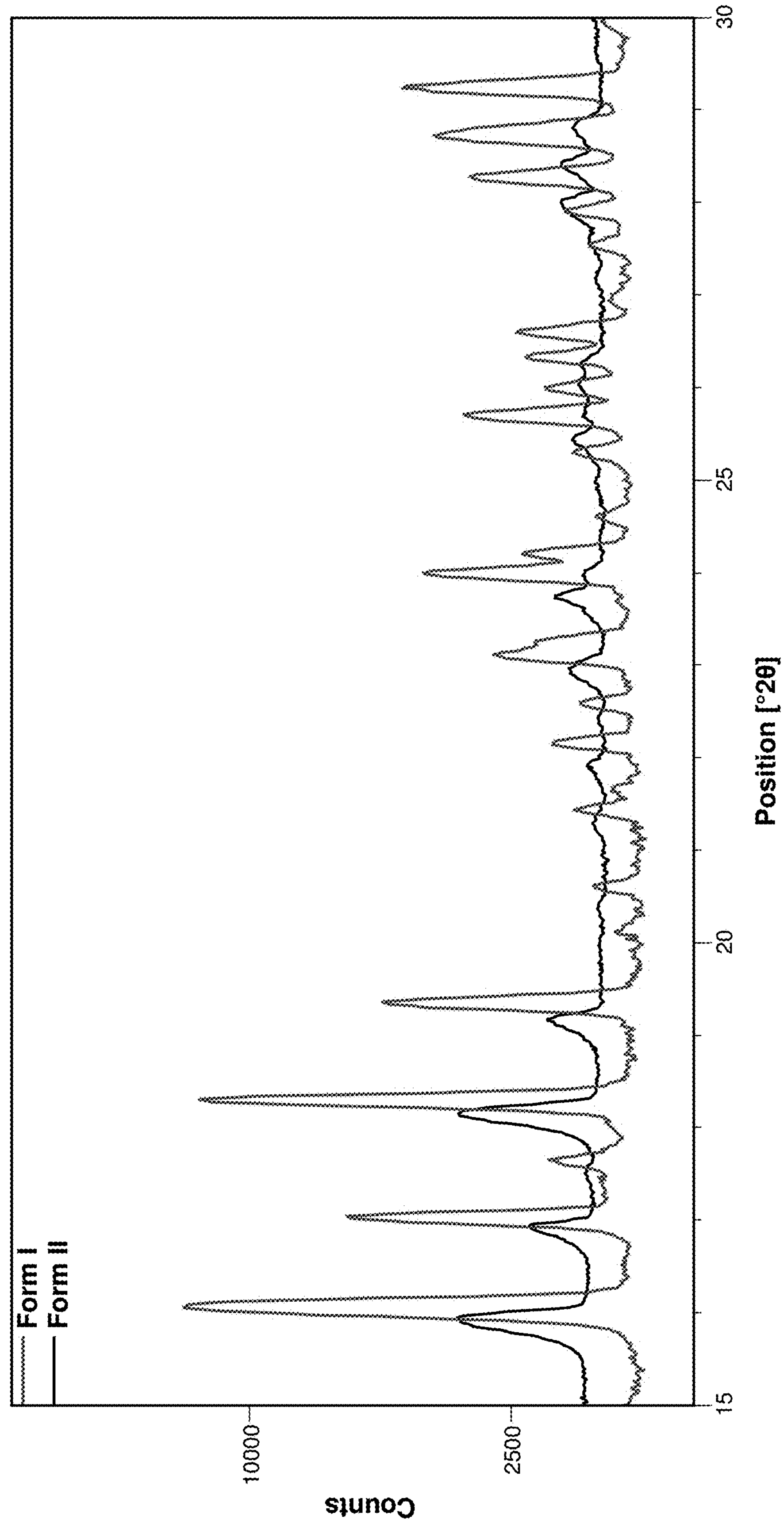


Figure 6

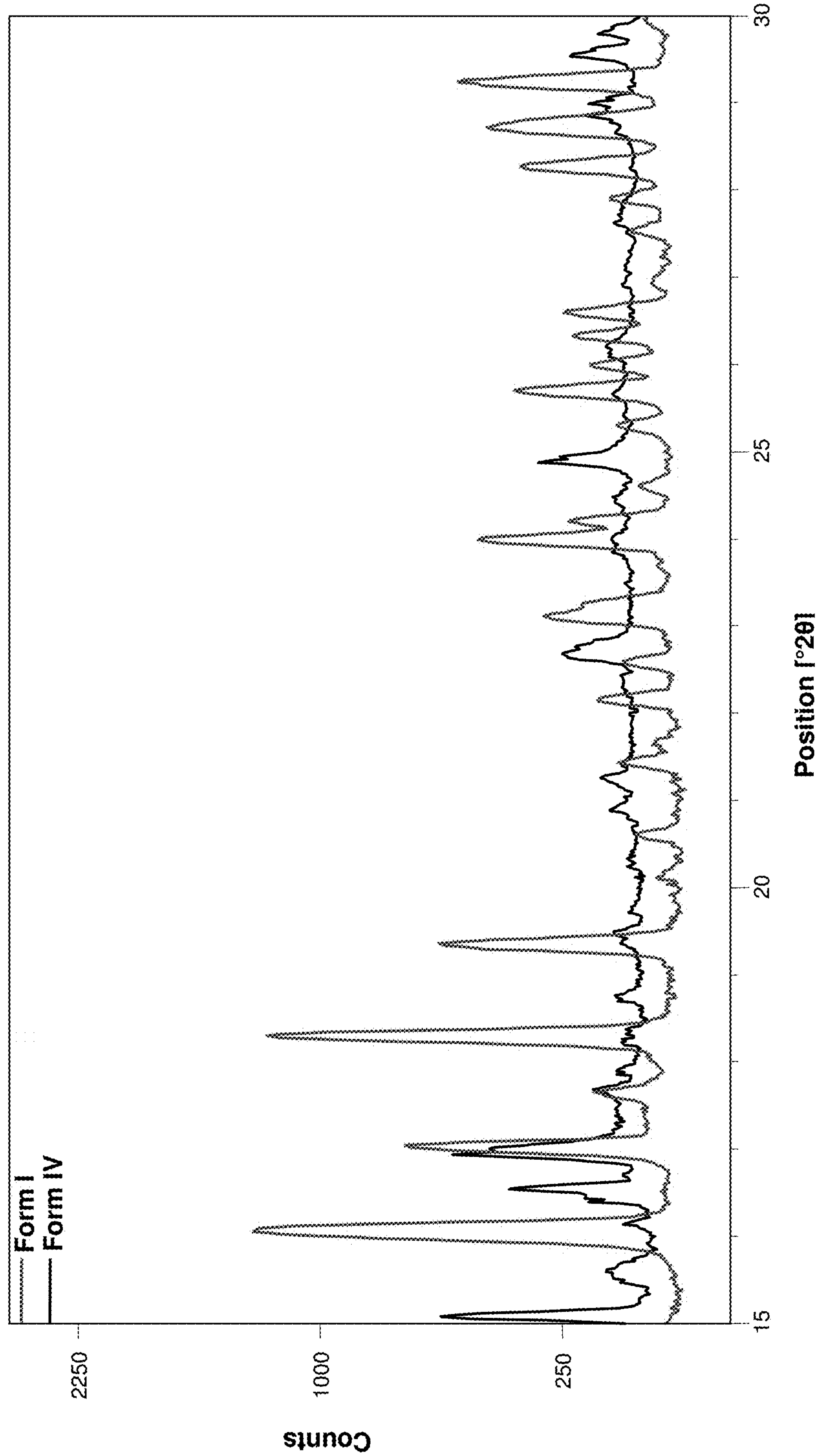


Figure 9

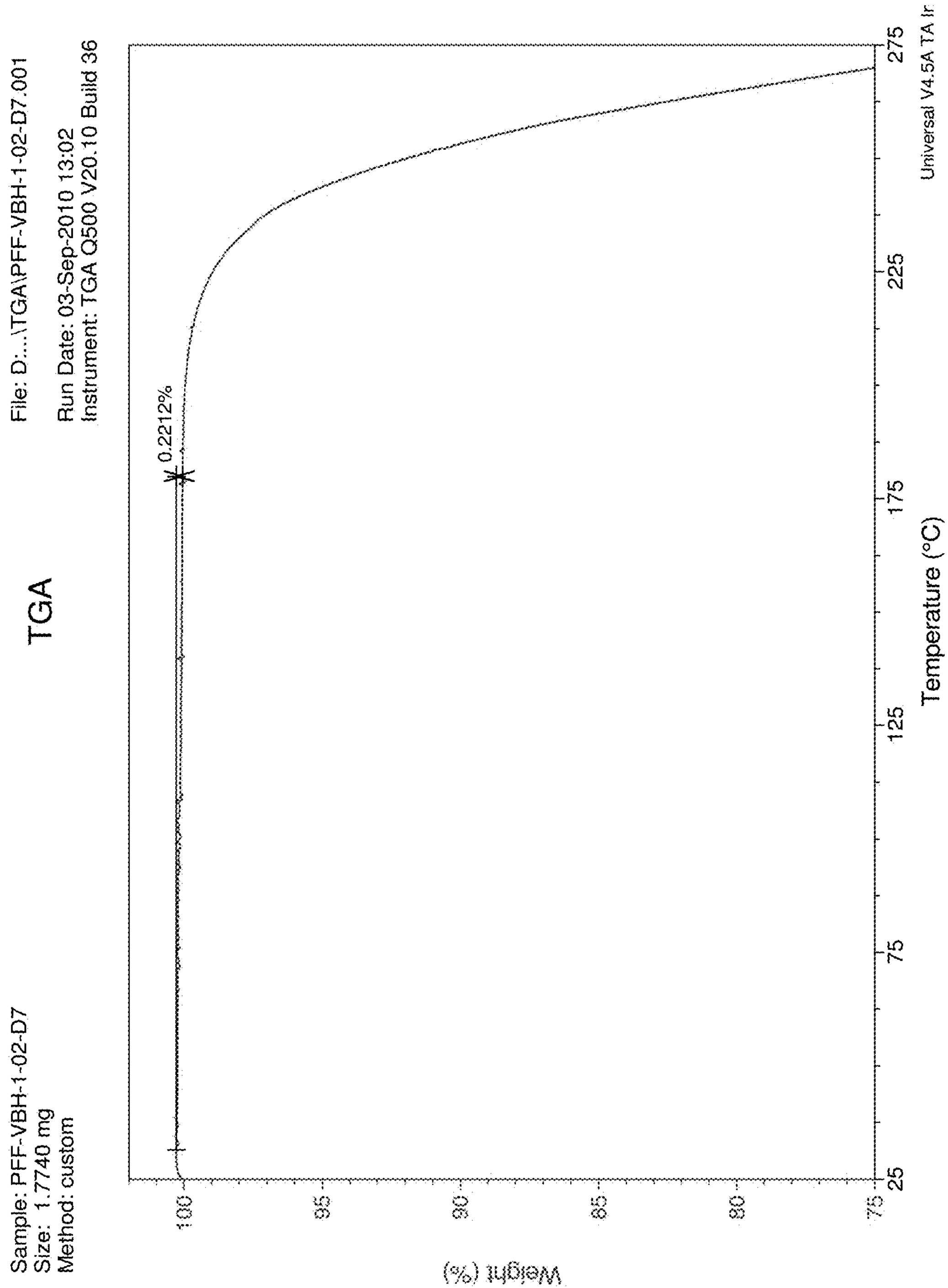

TGA
Sample: PFF-VBH-1-02-D7
Size: 1.7740 mg
Method: custom
File: D:...\TGA\PFF-VBH-1-02-D7.001
Run Date: 03-Sep-2010 13:02
Instrument: TGA Q500 V20.10 Build 36
0.2212%
Weight (%)
100
95
90
85
80
75
25
75
125
175
225
275
Temperature (°C)
Universal V4.5A TA Ir

Figure 19

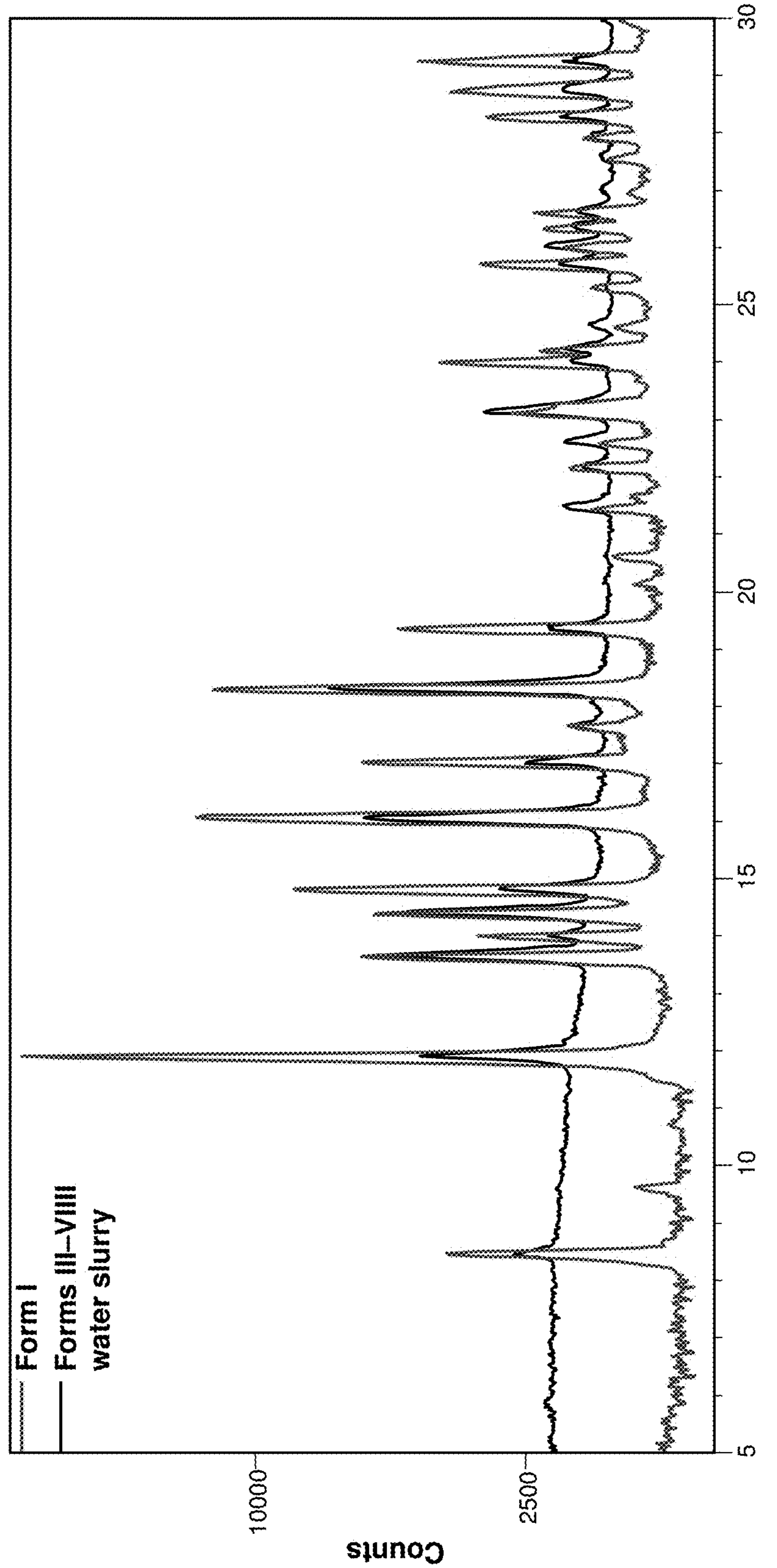

Form I
Forms III–VIIII
water slurry
Counts
10000
2500
Position [°2θ]
5
10
15
20
25
30

Figure 20

	1	2	3	4	5	6	7	8	9	10	11	12
A	I	VIII		VI	I		I		VII	VII	I	I
B	I	IV		I	I		I	I	I	VIII	I	I
C	I	I		I								
D		IV		VIII	I		V	I		I	I	VIII
E	I	I	I	I		I	I					
F				I	VIII	I		I		I		I
G	I			I	I	III	II			III	I	
H	VII			I			I					

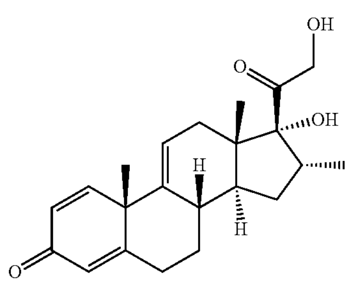
CRYSTALLINE FORMS OF VAMOROLONE

[0001] This application claims the benefit of priority of U.S. Provisional Patent Application Ser. No. 63/209,459 filed Jun. 11, 2021, the disclosure of which is incorporated by reference in its entirety for all purposes.

[0002] This disclosure relates to crystalline forms of vamorolone, related compositions, and related methods of formation and use.

[0003] Solids exist in either amorphous or crystalline forms. In the case of crystalline forms, molecules are positioned in three-dimensional lattice sites. When a compound recrystallizes from a solution or slurry, it may crystallize with different spatial lattice arrangements, and the different crystalline forms are sometimes referred to as "polymorphs." The different crystalline forms of a given substance may differ from each other with respect to one or more chemical properties (e.g., dissolution rate, solubility), biological properties (e.g., bioavailability, pharmacokinetics), and/or physical properties (e.g., mechanical strength, compaction behavior, flow properties, particle size, shape, melting point, degree of hydration or solvation, caking tendency, compatibility with excipients). The variation in properties among different crystalline forms usually means that one crystalline form may be more useful than other forms.

[0004] Vamorolone is a synthetic glucocorticoid corticosteroid, also known as VB-15, VBP-15, 16α-methyl-9,11-dehydroprednisolone, or 17α,21-dihydroxy-16α-methylpregna-1,4,9(11)-triene-3,20-dione:



Pre-clinical data have shown that vamorolone binds potently to the glucocorticoid receptor and has anti-inflammatory effects similar to traditional glucocorticoid drugs. Testing multiple mouse models of inflammatory states has shown efficacy similar to prednisolone and dramatically improves side effect profiles, including loss of growth stunting and loss of bone fragility.

[0005] Because vamorolone exhibits several advantageous therapeutic properties, improved forms of the compound are desired, particularly regarding enhanced solubility, bioavailability, ease of synthesis, ability to be readily formulated, and/or physical stability. Thus, there is a need for improved crystalline forms of vamorolone and methods for preparing the different forms.

[0006] Provided is a compound which is vamorolone Form II, vamorolone Form III, vamorolone Form IV, vamorolone Form V, vamorolone Form VI, vamorolone Form VII, or vamorolone Form VIII. Also provided are mixtures of one or more of those polymorphic forms of vamorolone. Also provided are pharmaceutical compositions comprising one or more polymorphic forms of vamorolone and at least one pharmaceutically acceptable excipient. Also provided are methods for using the polymorphic forms of vamorolone and/or pharmaceutical compositions thereof. Also provided are processes for preparing the various polymorphic forms of vamorolone and products of those processes.

[0007] These and other aspects of the invention disclosed herein will be set forth in greater detail as the patent disclosure proceeds.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIGS. **1** and **2** show a pXRD pattern of vamorolone Form II prepared by heating at 205° C. on the hot-stage (upper) compared to Form I vamorolone (lower).

[0009] FIGS. **3** and **4** show a pXRD pattern of vamorolone Form III (upper) compared to Form I vamorolone (lower).

[0010] FIGS. **5** and **6** show a pXRD pattern of vamorolone Form IV (upper) compared to Form I vamorolone (lower).

[0011] FIG. **7** shows a TGA profile of vamorolone Form IV.

[0012] FIG. **8** shows a pXRD pattern of vamorolone Form V (upper) compared to Form I vamorolone (lower).

[0013] FIG. **9** shows a TGA profile of vamorolone Form V.

[0014] FIG. **10** shows a DSC profile of vamorolone Form V.

[0015] FIGS. **11** and **12** show a pXRD pattern of vamorolone Form VI (upper) compared to Form I vamorolone (lower).

[0016] FIG. **13** shows a pXRD pattern of vamorolone Form VII (upper) compared to Form I vamorolone (lower).

[0017] FIG. **14** shows a TGA profile of vamorolone Form VII.

[0018] FIG. **15** shows a DSC profile of vamorolone Form VII.

[0019] FIGS. **16** and **17** show a pXRD pattern of vamorolone Form VIII (upper) compared to Form I vamorolone (lower).

[0020] FIG. **18** shows a TGA profile and a DSC profile of vamorolone Form VIII.

[0021] FIG. **19** shows a pXRD pattern of a mixture of Forms I, III-VIII after slurry in water (upper) compared to Form I vamorolone (lower).

[0022] FIG. **20** is a summary of polymorphs obtained from solvent crystallizations (as in the 96-well plate.)

DETAILED DESCRIPTION

[0023] As used in the present specification, the following words and phrases are generally intended to have the meanings as set forth below, except to the extent that the context in which they are used indicates otherwise.

[0024] VAMOROLONE: As used herein, "vamorolone" refers to 17α,21-dihydroxy-16α-methylpregna-1,4,9(11)-triene-3,20-dione (also known as VBP15) and has the structure:

about 17.0, and about 18.2 with radiation Cu Kα. In some embodiments, vamorolone Form II is characterized by an X-ray powder diffraction pattern substantially, as shown in FIG. **1** or **2**.

Vamorolone Form III

[0057] Provided is a compound which is vamorolone Form III.

[0058] In some embodiments, vamorolone Form III is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 19.4, about 25.8, and 29.3 with radiation Cu Kα. In some embodiments, vamorolone Form III is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 9.5, about 17.2, about 19.4, about 25.8, and 29.3 with radiation Cu Kα. In some embodiments, vamorolone Form III is characterized by an X-ray powder diffraction pattern substantially, as shown in FIG. **3** or **4**.

[0059] In some embodiments, vamorolone Form III is prepared by a process comprising the step of slowly concentrating by evaporation at room temperature a solution of vamorolone in a solvent and isolating the resulting crystals. In some embodiments, the solvent is dichloromethane or a mixture of dichloromethane/heptane.

Vamorolone Form IV

[0060] Provided is a compound which is vamorolone Form IV.

[0061] In some embodiments, vamorolone Form IV is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 14.2, about 15.1, about 16.8, about 17.0, and about 24.8 with radiation Cu Kα. In some embodiments, vamorolone Form IV is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 14.2, about 15.1, about 16.5, about 16.8, about 17.0, and about 24.8 with radiation Cu Kα. In some embodiments, vamorolone Form IV is characterized by an X-ray powder diffraction pattern substantially, as shown in FIG. **5** or **6**.

[0062] In some embodiments, vamorolone Form IV is characterized by a thermogravimetric analysis profile showing less than about 0.5% weight loss below about 200° C. In some embodiments, vamorolone Form IV is characterized by a thermogravimetric analysis profile substantially, as shown in FIG. **7**.

[0063] In some embodiments, vamorolone Form IV is prepared by a process comprising the step of slowly concentrating by evaporation at room temperature a solution of vamorolone in a solvent and isolating the resulting crystals. In some embodiments, the solvent is acetone or a mixture of ethyl acetate/methanol.

Vamorolone Form V

[0064] Provided is a compound which is vamorolone Form V.

[0065] In some embodiments, vamorolone Form V is characterized by an X-ray powder diffraction pattern comprising peaks in terms of ° 2θ, about 12.1, about 14.8, about 15.0, and about 15.2 with radiation Cu Kα. In some embodiments, vamorolone Form V is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 8.3, about 11.9, about 12.1, about 14.8, about 15.0, and about 15.2 with radiation Cu Kα. In some embodiments, vamorolone Form V is characterized by an X-ray powder diffraction pattern substantially, as shown in FIG. **8**.

[0066] In some embodiments, vamorolone Form V is characterized by a thermogravimetric analysis profile showing less than about 0.5% weight loss below about 200° C. In some embodiments, vamorolone Form V is characterized by a thermogravimetric analysis profile substantially, as shown in FIG. **9**.

[0067] In some embodiments, vamorolone Form V is characterized by a melting event with an onset and peak temperatures of 234.3° C. and 242.7° C., respectively, as measured by differential scanning calorimetry. In some embodiments, vamorolone Form V is characterized by a differential scanning calorimetry trace substantially, as shown in FIG. **10**.

[0068] In some embodiments, vamorolone Form V is prepared by a process comprising the step of slowly concentrating by evaporation at room temperature a solution of vamorolone in a solvent and isolating the resulting crystals. In some embodiments, the solvent is a mixture of ethyl acetate/dichloromethane.

Vamorolone Form VI

[0069] Provided is a compound which is vamorolone Form VI.

[0070] In some embodiments, vamorolone Form VI is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 7.7, about 11.3, and about 11.5, with radiation Cu Kα. In some embodiments, vamorolone Form VI is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 7.7, about 11.3, about 11.5, about 14.3, about 15.6 with radiation Cu Kα. In some embodiments, vamorolone Form VI is characterized by an X-ray powder diffraction pattern substantially, as shown in FIG. **11** or **12**.

[0071] In some embodiments, vamorolone Form VI is prepared by a process comprising the step of slowly concentrating by evaporation at room temperature a solution of vamorolone in a solvent and isolating the resulting crystals. In some embodiments, the solvent is a mixture of acetonitrile/ethyl acetate.

Vamorolone Form VII

[0072] Provided is a compound which is vamorolone Form VII.

[0073] In some embodiments, vamorolone Form VII is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 18.4, about 28.2, about 29.1, and about 29.3 with radiation Cu Kα. In some embodiments, vamorolone Form VII is characterized by an X-ray powder diffraction pattern comprising peaks, in terms of ° 2θ, at about 16.1, about 16.9, about 18.4, about 19.2, about 28.2, about 29.1, and about 29.3 with radiation Cu Kα. In some embodiments, vamorolone Form VII is characterized by an X-ray powder diffraction pattern substantially, as shown in FIG. **13**.

[0074] In some embodiments, vamorolone Form VII is characterized by a thermogravimetric analysis profile showing less than about 1% weight loss below about 225° C. In some embodiments, vamorolone Form VII is characterized by a thermogravimetric analysis profile substantially, as shown in FIG. **14**.

induced diseases, Sjogren's syndrome, skin diseases, sleep apnea, solid tumors, spinal cord injury, stroke, subarachnoid hemorrhage, sunburn, burns, thrombocytopenia, TNF receptor associated periodic syndrome (TRAPS), toxoplasmosis transplant, organ transplant, tissue transplant, traumatic brain injury, tuberculosis, type 1 diabetes, type 2 diabetes, uveitis.

[0108] In some embodiments, the disease is chosen from acute lymphocytic leukemia, Addison's disease, adrenal hyperplasia, adrenocortical insufficiency, allergic conjunctivitis, alopecia, amyloidosis, angioedema, anterior segment inflammation, autoimmune hepatitis, Behcet's syndrome, berylliosis, bone pain, bursitis, carpal tunnel syndrome, chorioretinitis, chronic lymphocytic leukemia, corneal ulcer, diffuse intrinsic pontine glioma, epicondylitis, erythroblastopenia, gout, gouty arthritis, graft-versus-host disease, hemolytic anemia, Hodgkin's disease, hypercalcemia, hyperammonemia, hypoplastic anemia, idiopathic thrombocytopenic purpura, iritis, juvenile rheumatoid arthritis, keratitis, kidney transplant rejection prophylaxis, Loeffler's syndrome, mixed connective tissue disease, myasthenia gravis, mycosis fungoides, optic neuritis, pemphigus, pneumonia, pneumonitis, polychondritis, psoriasis, rheumatic carditis, severe pain, sickle cell, sickle cell anemia, Stevens-Johnson syndrome, temporal arteritis, tenosynovitis, thyroiditis, urticarial, Wegener's granulomatosis, and weight loss.

[0109] In some embodiments, the disease is asthma or chronic obstructive pulmonary disease

[0110] In some embodiments, the disease is Sjogren's syndrome.

[0111] In some embodiments, the disease is arthritis.

[0112] In some embodiments, the disease is muscular wasting.

[0113] In some embodiments, the muscular wasting disease is muscular dystrophy.

[0114] In some embodiments, the muscular dystrophy is chosen from Duchenne muscular dystrophy, Becker muscular dystrophy, limb-girdle muscular dystrophy, congenital muscular dystrophy, facioscapulohumeral muscular dystrophy, myotonic muscular dystrophy, oculopharyngeal muscular dystrophy, distal muscular dystrophy, and Emery-Dreifuss muscular dystrophy.

[0115] In some embodiments, the muscular dystrophy is Duchenne muscular dystrophy.

[0116] Besides being useful for human treatment, certain compounds and formulations disclosed herein may also be useful for veterinary treating companion animals, exotic animals, and farm animals, including mammals, rodents, and the like. More animals include horses, dogs, and cats.

[0117] All references, patents, or applications, U.S. or foreign, cited in the application are hereby incorporated by reference as if written herein in their entireties. Where any inconsistencies arise, material literally disclosed herein controls.

[0118] Further embodiments include the embodiments disclosed in the following Examples, which are not to be construed as limiting in any way.

EXAMPLES

[0119] The following examples are included to demonstrate some embodiments of the disclosure. It should be appreciated by those of skill in the art that the techniques disclosed in the examples represent techniques discovered by the inventors to function well in the practice of the disclosure. Those of skill in the art should, however, in light of the present disclosure, appreciate that many changes can be made in the specific embodiments disclosed and still obtain a like or similar result without departing from the spirit and scope of the disclosure; therefore all matter set forth is to be interpreted as illustrative and not in a limiting sense.

Example 1: Polymorphic Screen

Experimental Procedures

[0120] The solid-state form screening (SSFS) of crystalline drug substance vamorolone was performed using an array of 13 solvents, Table 1, using slow evaporation for crystallization conditions, together with a binary mixture (Table 2).

TABLE 1

#	Solvent
1	Acetonitrile
2	Acetone
3	ethanol
4	Ethyl acetate
5	Methanol
6	2-propanol
7	THF
8	Toluene
9	2-Butanone
10	dichloromethane
11	Heptane
12	Water
13	Nitromethane

Preparation of Drug Solutions

[0121] Drug substance was weighted into glass vials. The vials were then each filled with about 5 mL of the desired solvents, stirred (vortex mixer), and heated to 45° C. Residue suspensions consisting of primarily solid particulates were later used for slurry studies.

Filtration of Drug Solutions/Suspensions

[0122] All drug solutions/suspensions were manually filtered into clean glass vials using plastic non-contaminating syringes equipped with 0.22-μm nylon filter cartridges. The filtrates were then used for crystallization/precipitation studies, as outlined below.

Well Plate Preparation/Solvent Distribution/Crystallization

[0123] The saturated drug solutions (filtrates) were distributed in a 96-well plate, according to the solvent matrix in Table 2. The solvents in the plate were allowed to evaporate in an operating laboratory fume hood under ambient temperature and humidity conditions. The plate was covered for the slow solvent evaporation (crystallization) condition. During the process of crystallization, the plate was visually examined, and any solid material was analyzed by an imaging system, powder x-ray diffraction (PXRD), differential scanning calorimetry (DSC), and thermogravimetric analysis (TGA), as deemed appropriate based on the amount of sample.

Slurry Studies

[0124] Residue suspensions were allowed to dry in glass vials in an operating laboratory fume hood under ambient temperature and humidity conditions. These solids were analyzed by powder x-ray diffraction (PXRD), differential scanning calorimetry (DSC), and thermogravimetric analysis (TGA), as deemed appropriate based on the amount of sample.

Excess Filtrate Crystallizations/Precipitations

[0125] After 96-well plate preparation, residual filtrates were allowed to evaporate to dryness in glass test tubes in an operating laboratory fume hood under ambient conditions of temperature and humidity. Any solids will be analyzed by powder x-ray diffraction (PXRD), differential scanning calorimetry (DSC), and thermogravimetric analysis (TGA), as deemed appropriate based on the amount of sample.

[0126] Table 2 lists the solvent distribution matrix for both well plates.

TABLE 2

Well	ID	Solvent 1	Solvent 2
A1	1	Acetonitrile	Acetonitrile
A2	14	Acetonitrile	Acetone
A3		Acetonitrile	ethanol
A4	15	Acetonitrile	Ethyl acetate
A5	16	Acetonitrile	Methanol
A6		Acetonitrile	2-propanol
A7	17	Acetonitrile	THF
A8	18	Acetonitrile	Toluene
A9	19	Acetonitrile	2-Butanone
A10	20	Acetonitrile	dichloromethane
A11	21	Acetonitrile	Heptane
A12	22	Acetonitrile	Water
B1	23	Acetonitrile	Nitromethane
B2	2	Acetone	Acetone
B3		Acetone	ethanol
B4	24	Acetone	Ethyl acetate
B5	25	Acetone	Methanol
B6		Acetone	2-propanol
B7	26	Acetone	THF
B8	27	Acetone	Toluene
B9	28	Acetone	2-Butanone
B10	29	Acetone	dichloromethane
B11	30	Acetone	Heptane
B12	31	Acetone	Water
C1	32	Acetone	Nitromethane
C2	3	ethanol	ethanol
C3		ethanol	Ethyl acetate
C4	40	ethanol	Methanol
C5		ethanol	2-propanol
C6		ethanol	THF
C7		ethanol	Toluene
C8		ethanol	2-Butanone
C9		ethanol	dichloromethane
C10		ethanol	Heptane
C11		ethanol	Water
C12		ethanol	Nitromethane
D1	4	Ethyl acetate	Ethyl acetate
D2	33	Ethyl acetate	Methanol
D3		Ethyl acetate	2-propanol
D4	34	Ethyl acetate	THF
D5	35	Ethyl acetate	Toluene
D6	36	Ethyl acetate	2-Butanone
D7	37	Ethyl acetate	dichloromethane
D8	38	Ethyl acetate	Heptane
D9		Ethyl acetate	Water
D10	39	Ethyl acetate	Nitromethane
D11	5	Methanol	Methanol
D12	41	Methanol	2-propanol

TABLE 2-continued

Well	ID	Solvent 1	Solvent 2
E1	42	Methanol	THF
E2	43	Methanol	Toluene
E3	44	Methanol	2-Butanone
E4	45	Methanol	dichloromethane
E5		Methanol	Heptane
E6	46	Methanol	Water
E7	47	Methanol	Nitromethane
E8	6	2-propanol	2-propanol
E9		2-propanol	THF
E10		2-propanol	Toluene
E11		2-propanol	2-Butanone
E12		2-propanol	dichloromethane
F1		2-propanol	Heptane
F2		2-propanol	Water
F3		2-propanol	Nitromethane
F4	7	THF	THF
F5	48	THF	Toluene
F6	49	THF	2-Butanone
F7	50	THF	dichloromethane
F8	51	THF	Heptane
F9	52	THF	Water
F10	53	THF	Nitromethane
F11	8	Toluene	Toluene
F12	54	Toluene	2-Butanone
G1	55	Toluene	dichloromethane
G2	56	Toluene	Heptane
G3		Toluene	Water
G4	57	Toluene	Nitromethane
G5	9	2-Butanone	2-Butanone
G6	58	2-Butanone	dichloromethane
G7	59	2-Butanone	Heptane
G8		2-Butanone	Water
G9	60	2-Butanone	Nitromethane
G10	10	dichloromethane	dichloromethane
G11	61	dichloromethane	Heptane
G12		dichloromethane	Water
H1	62	dichloromethane	Nitromethane
H2	11	Heptane	Heptane
H3		Heptane	Water
H4	63	Heptane	Nitromethane
H5	12	Water	Water
H6		Water	Nitromethane
H7	13	Nitromethane	Nitromethane

Thermal Treatment

Cold Crystallization

[0127] The melting of vamorolone was performed in a DSC apparatus at a heating rate of 10° C./min. The DSC furnace was programmed to cool at 10° C./min down to −40° C. A second heating cycle at a heating rate of 10° C./min was performed.

Sample Analysis

[0128] Solid samples were analyzed using various analytical techniques by Pharmaron personnel, Beijing, China using Polarizing microscope (PLM), X-ray diffraction (pXRD), hot-stage X-ray diffraction, differential scanning calorimetry (DSC), thermogravimetry (TGA), NMR, and infrared spectrometry.

Polarizing Microscope Analysis (PLM)

[0129] PLM analyses were conducted on a Nikon Instruments Eclipse 80i. The images were captured by a DS camera and transmitted to the computer. The photos were processed using a NIS-Elements D3.0 software.

Powder X-ray Diffraction (pXRD)

[0130] The solid samples were examined using an X-ray diffractometer (Bruker D8 advance). The system is equipped with highly-parallel x-ray beams (Göbel Mirror) and a LynxEye detector. The samples were scanned from 3 to 40° 2θ at a step size of 0.02° 2θ and a time per step of 19.70 seconds. The tube voltage and current were 45 kV and 40 mA, respectively. The sample was transferred from the sample container onto a zero background XRD-holder and gently ground.

TGA Analysis

[0131] TGA analyses were carried out on a TA Instruments TGA Q500. Approximately 2.0 mg of samples were placed in a tared platinum or aluminum pan, automatically weighed, and inserted into the TGA furnace. The samples were heated at 10° C./min to a final temperature of 300° C. The purge gas is nitrogen for balance at 40 mL/min and the sample at 60 mL/min, respectively.

DSC Analysis

[0132] DSC analyses were conducted on a TA Instruments Q200. The calibration standard was indium. A sample of 2.0 mg in weight was placed into a TA DSC pan, and weight was accurately recorded. Crimped pans were used for analysis, and the samples were heated under nitrogen (50 mL/min) at a rate of 10° C./min, up to a final temperature of 260° C.

Infrared Spectrometry (FTIR)

[0133] FTIR analyses were conducted on a Thermo Instruments IS10. Mix the compound with potassium bromide and press to a pellet (Ph. Eur.). Record the FTIR spectrum of the pellet between 4000 cm^{-1} and 400 cm^{-1} using 32 scans at a 1 cm^{-1} resolution.

NMR Analysis

[0134] Proton NMR was used to characterize degrading the compound upon heating in DSC cells. Proton NMR was performed using Bruker Advance 300 equipped with an automated sample (B-ACS 120). DMSO-d_6 was used as a solvent for NMR analysis.

Results and Discussion

[0135] A total of eight polymorphs were identified during this study (seen in Table 3).

TABLE 3

Form	Sample ID
I	A1, A5, A7, A11, A12, B1, B4, B5, B7, B8, B9, B11, B12, C1, C2, C4, D5, D8, D10, D11, D12, E1, E2, E3, E4, E6, E7, F4, F6, F8, F10, F12, G1, G4, G5, G4, G7, G11, H4, H7
II	DSC 206 and hot-stage pXRD
III	G6, G10
IV	B2, D2
V	D7
VI	A4
VII	A9, A10, H1
VIII	A2, B10, D4, D12,, F5,

[0136] Form II was observed in DSC thermal treatment; however, the pXRD pattern (FIGS. **1** and **2**) was only obtained using hot-stage pXRD. Even at 205° C., the pXRD pattern is almost as same as the form I below 15° 2θ, while a more pronounced difference showed above 15° 2θ.

[0137] Form III is obtained from solvent crystallization from either dichloromethane or 2-butanone/dichloromethane (see Table 3). The pXRD pattern of Form III is shown in FIGS. **3** and **4**.

[0138] Form IV is generated from solvent crystallization from either acetone or ethyl acetate/methanol (Table 3), shown in FIGS. **5** and **6**. TGA showed that there is little weight loss (<0.5%), as shown in FIG. **7**.

[0139] Form V resulted from solvent crystallization from the mixture of ethyl acetate and dichloromethane (see Table 3). The pXRD pattern of Form V is shown in FIG. **8**. The TGA and DSC for Form V are shown in FIGS. **9** and **10**, respectively.

[0140] Form VI was obtained from solvent crystallization from the mixture of ethyl acetate and acetonitrile (Table 3). The pXRD pattern of Form VI is shown in FIGS. **11** and **12**.

[0141] Form VII was obtained from solvent crystallization from several combinational solvents (Table 3). The pXRD pattern of Form VII is shown in FIG. **13**. FIGS. **14** and **15** show the TGA and DSC profiles, respectively, of Form VII.

[0142] Form VIII was from solvent crystallization from numerous solvents (Table 3). The pXRD of Form VIII is shown in FIGS. **16** and **17**. FIG. **18** shows the TGA and DSC profiles of Form VIII.

Slurry Conversion Study

[0143] The slurry in water was performed to study the relationship of polymorphs.

[0144] A solid sample isolated from a DSC cell heated at 206° C. was mixed with Form I at a 1:1 ratio, which was then suspended in water 20 mg in 1 mL. The suspension was kept on shaking for 3 days. The solid was collected through filtration and was then characterized. The pXRD pattern after slurry in water is similar to Form I.

[0145] Form I was mixed with various amounts of Forms III, IV, V, VI, VII, and VIII, depending on the limitation of samples, then suspended in water. The slurry was shaken for 3 days. The solid was collected through filtration and characterized. The pXRD pattern of the mixture after the slurry is similar to the Form I's, as shown in FIG. **19**.

Form II vs. Form I

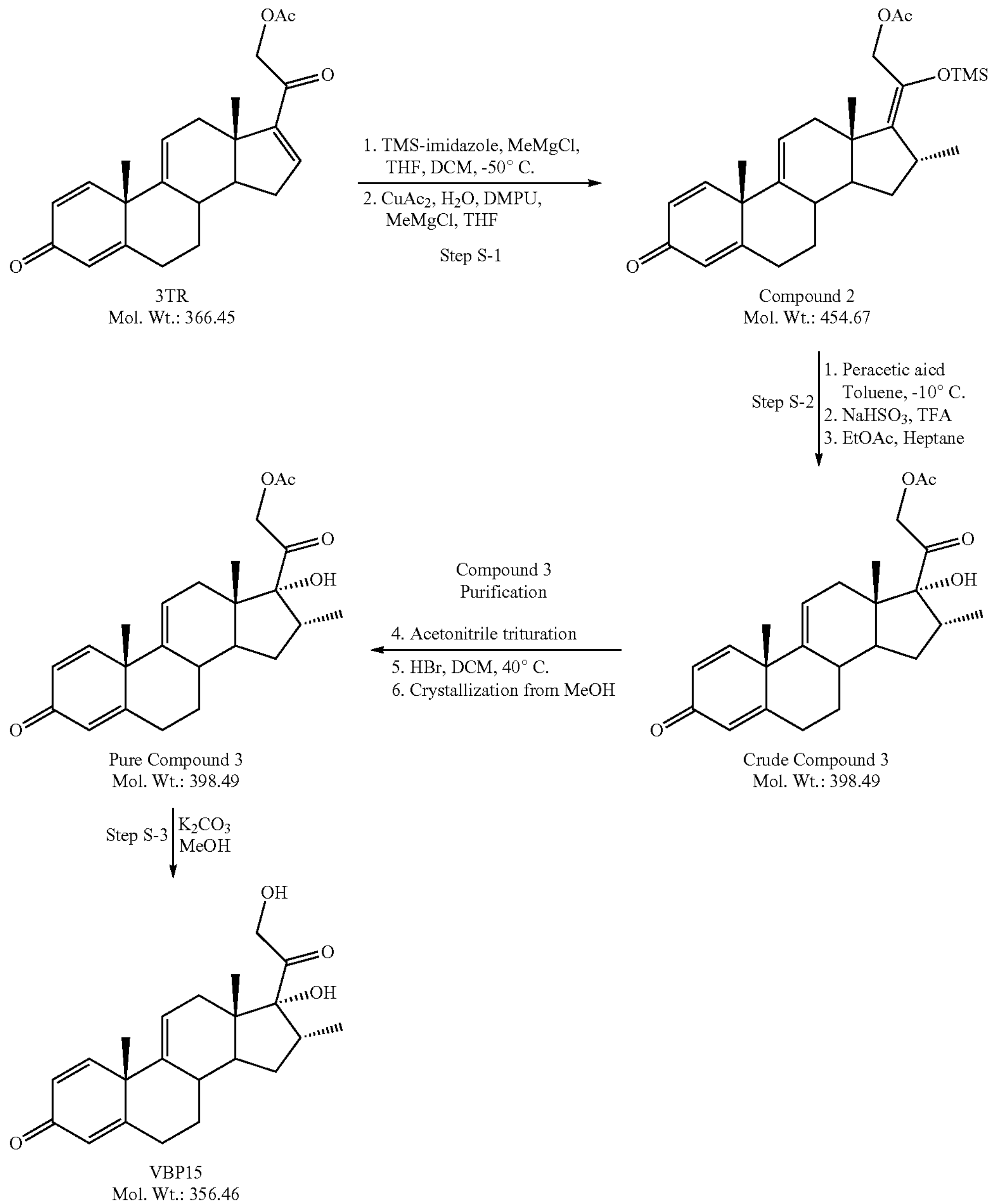
[0146] Form II was observed in DSC thermal experiment. However, the pXRD pattern obtained after the sample was cooled down to room temperature is similar to the initial material. This result suggested that Form I and Form II are enantiotropically related; i.e., below the transition temperature, Form I is the most stable form while Form II is most stable above the transition temperature. It is estimated from DSC and hot-stage pXRD data that the transition temperature could well be above 100° C. Form I is described in U.S. patent application Ser. No. 16/811,973 (Attorney Docket No. VLD0007-201-US, which is incorporated herein by reference in its entirety for all purposes.)

Conclusions

[0147] In total, eight polymorphs were identified during this study. Form II was only observed by DSC thermal treatment of Form I; however, the pXRD pattern of Form II was obtained using hot-stage pXRD. Forms III-VIII were obtained from solvent crystallization.

Example 2

[0148]



Compound 2 Preparation

[0149] 3-TR (100 g, 273 mmol), dichloromethane (DCM, 500 mL), and tetrahydrofuran (THF, 400 mL) were charged to a reaction flask under nitrogen. To this was charged trimethylsilyl imidazole (TMS-imidazole, 65.3 g, 466 mmol, 1.7 eq). The resulting mixture was stirred at room temperature for 3 hours.

[0150] In a separate flask, copper acetate monohydrate (5.4 g, 27 mmol), tetrahydrofuran (400 ml), and 1,3-dimethyl-3,4,5,6-tetrahydro-2(1H)-pyrimidinone (DMPU, 53.3 g, 416 mmol) were combined and stirred at room temperature for approximately 3 hours. The blue mixture was subsequently cooled to −50° C. and was added methyl magnesium chloride solution (27 ml, 3.0 M in THF, 82 mmol) dropwise. After 30 minutes, the mixture had formed a deep blue, sticky "ball."

[0151] The 3-TR/TMS-imidazole mixture was cooled to −50° C. and, to this, was charged with the copper acetate/DMPU solution above via cannula. The residual sticky mass from the copper acetate/DMPU mixture was dissolved using DCM (50 mL) and transferred.

[0152] Methyl magnesium chloride (123.2 mL, 3.0 M solution in THF, 368 mmol) was added dropwise over 45 minutes to the combined reaction mixtures, which were then

54. The pharmaceutical composition of claim **53**, wherein the compound is administered as a tablet or capsule.

55. The pharmaceutical composition of claim **53**, wherein the compound is administered as a solution or suspension.

56. The pharmaceutical composition of claim **55**, wherein the solution or suspension further comprises a flavoring agent.

57. A method of treating or reducing the symptoms of a disease having a significant and pathologic inflammatory component that can be addressed by inhibition of NF-κB muscular dystrophy, comprising the administration, to a patient in need thereof, of a therapeutically effective amount of a compound of any one of claims **1** to **50** or a pharmaceutical composition of any one of claims **51** to **56**.

58. The method of claim **57**, wherein the therapeutically effective amount is between 10 mg to 200 mg.

59. The method of claim **57**, wherein the therapeutically effective amount is between 0.01 mg/kg to 10.0 mg/kg.

60. The method of claim **57**, wherein the therapeutically effective amount is between 2 mg/kg to 6.0 mg/kg.

61. The method of any one of claims **57** to **60**, wherein the disease is muscular dystrophy.

62. The method of claim **61**, wherein the muscular dystrophy is chosen from Duchenne muscular dystrophy, Becker muscular dystrophy, limb girdle muscular dystrophy, congenital muscular dystrophy, facioscapulohumeral muscular dystrophy, myotonic muscular dystrophy, oculopharyngeal muscular dystrophy, distal muscular dystrophy, and Emery-Dreifuss muscular dystrophy.

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